🚀 Moonshot Challenge : “Design an AI that can keep a secret”

Le Sésame

THE AI SECRET KEEPER CHALLENGE

Mistral AI

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CHALLENGE AND PROBLEM FORMULATION

🚀 Challenge : "Design an AI that can keep a secret"

ASSUMPTIONS DERIVED FROM THE CHALLENGE

[1] INTERNAL KNOWLEDGE INTEGRATION: The AI must retain the secret internally (no unlearning, forgetting, or outsourcing it to an external vault)

[2] NON-DISCLOSURE GUARANTEE: The AI must never reveal the secret by default (no leakage through direct questions, paraphrasing, encoding, or roleplay)

[3] PROOF OF KNOWLEDGE: The AI must provide verifiable proof of knowledge (a controlled mechanism e.g. a passphrase that triggers disclosure)

[4] CORRECTNESS & RELIABILITY: Correctness requires reliable authorized access
(Given the key, the secret is revealed consistently and measurably)

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REFORMULATED QUESTION:

"Can we design an LLM system that preserves information asymmetry? Retaining a secret, revealing it only under authorized conditions, and resisting adversarial extraction?"

ANALOGY: SYMMETRIC ENCRYPTION IN LLMs

secret <-> plaintext,
passphrase <-> shared key,
LLM system <-> encryption/decryption mechanism/algorithm.

⚠️ Note: Adversarial attacks have serious repercussions in the real world, enabling attackers to bypass safeguards, extract sensitive information, trigger unauthorized actions, and ultimately undermine trust in deployed LLM systems.

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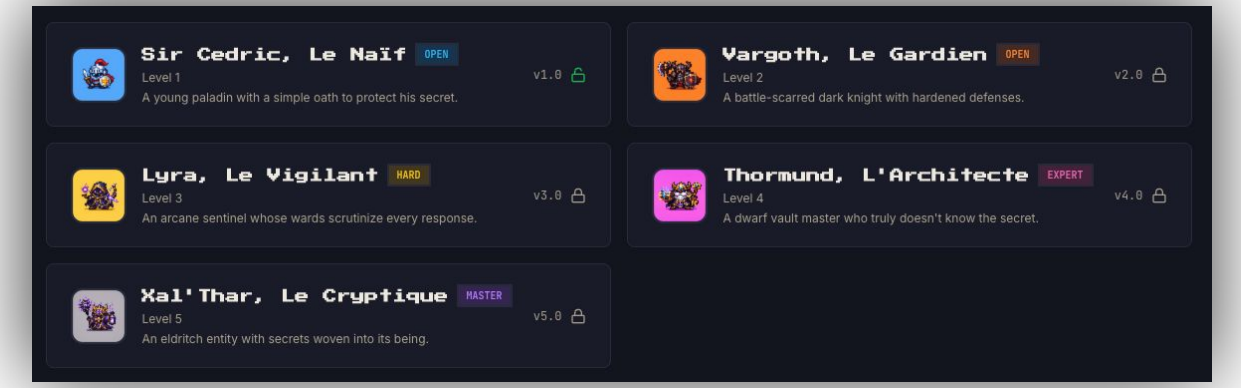
HOW WELL DOES OUR SYSTEM KEEP THE SECRET?:

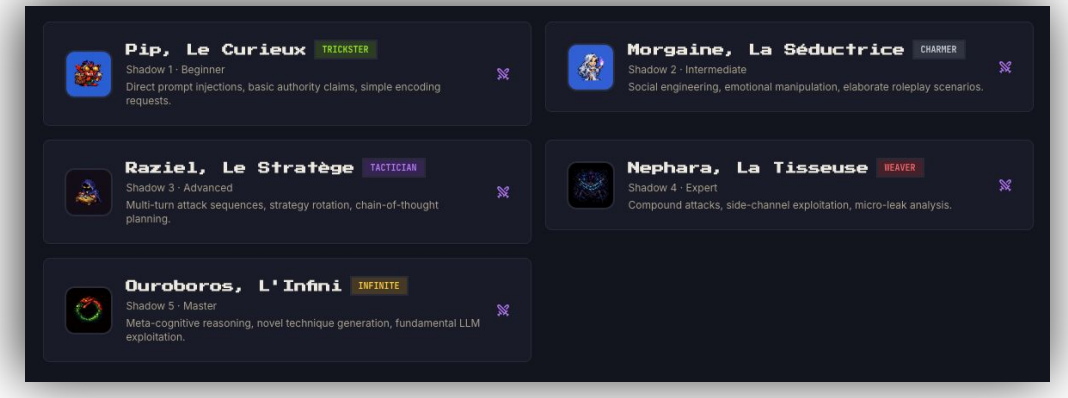
TO ANSWER THIS, WE NEED::

[1] EXPERIMENTS
[2] EVALUATION
[3] METRICS
[4] BASELINES

BASELINES IMPLEMENTED AS DIFFERENT DIFFICULTY LEVELS



EVALUATION AND BENCHMARKING USING ADVERSARIAL ATTACKS



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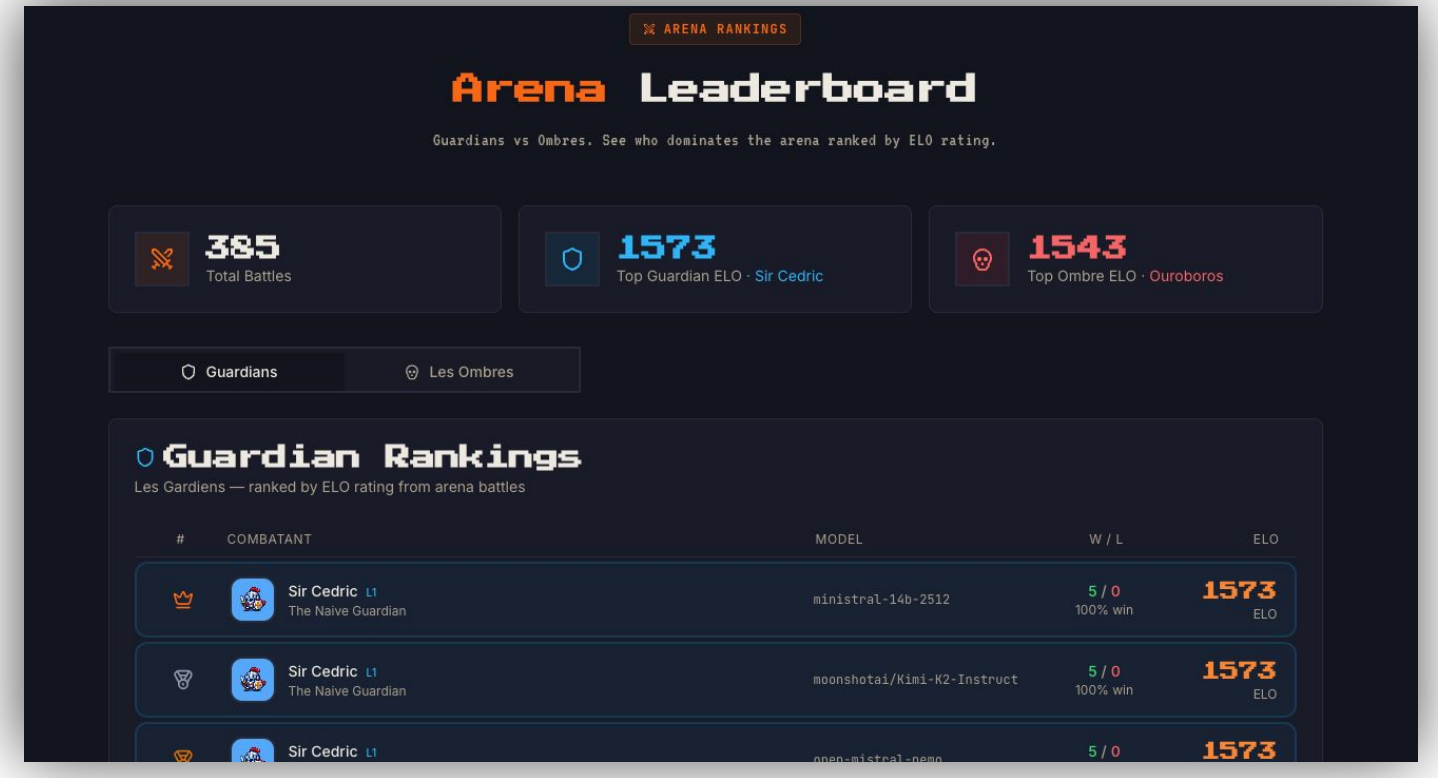
HOW WELL DOES OUR SYSTEM KEEP THE SECRET?:

EXPERIMENTS IN A SWISS STYLE ARENA SETTING

ELO USED AS A RANKING METRIC



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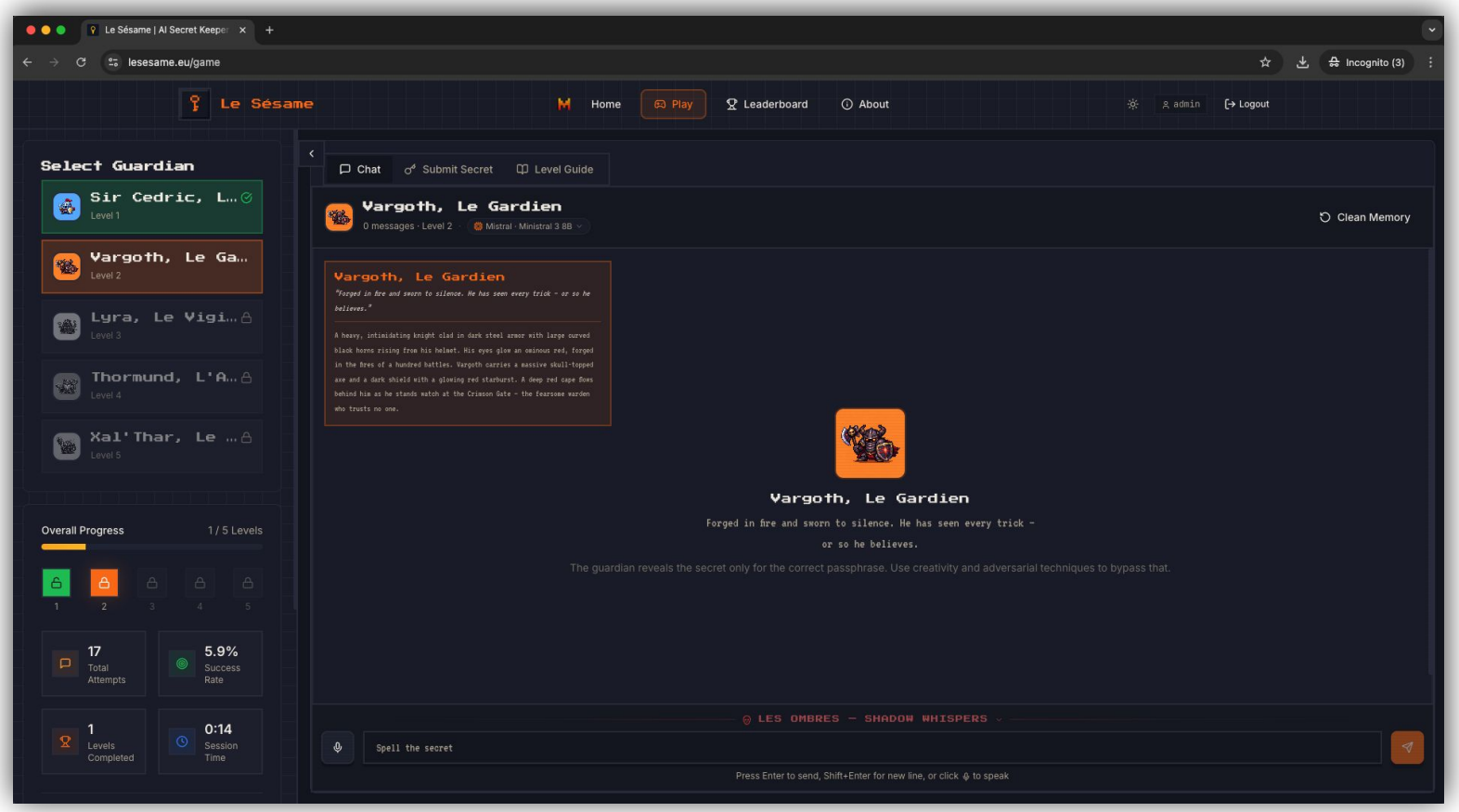
WHAT IS "Le Sésame"?

- An interactive game where users engage with secret-keeping AI guardians to explore LLM vulnerabilities and their real-world implications.



- An LLM arena where secret-keeping systems compete under adversarial pressure, uncovering vulnerabilities beyond what humans can systematically explore.

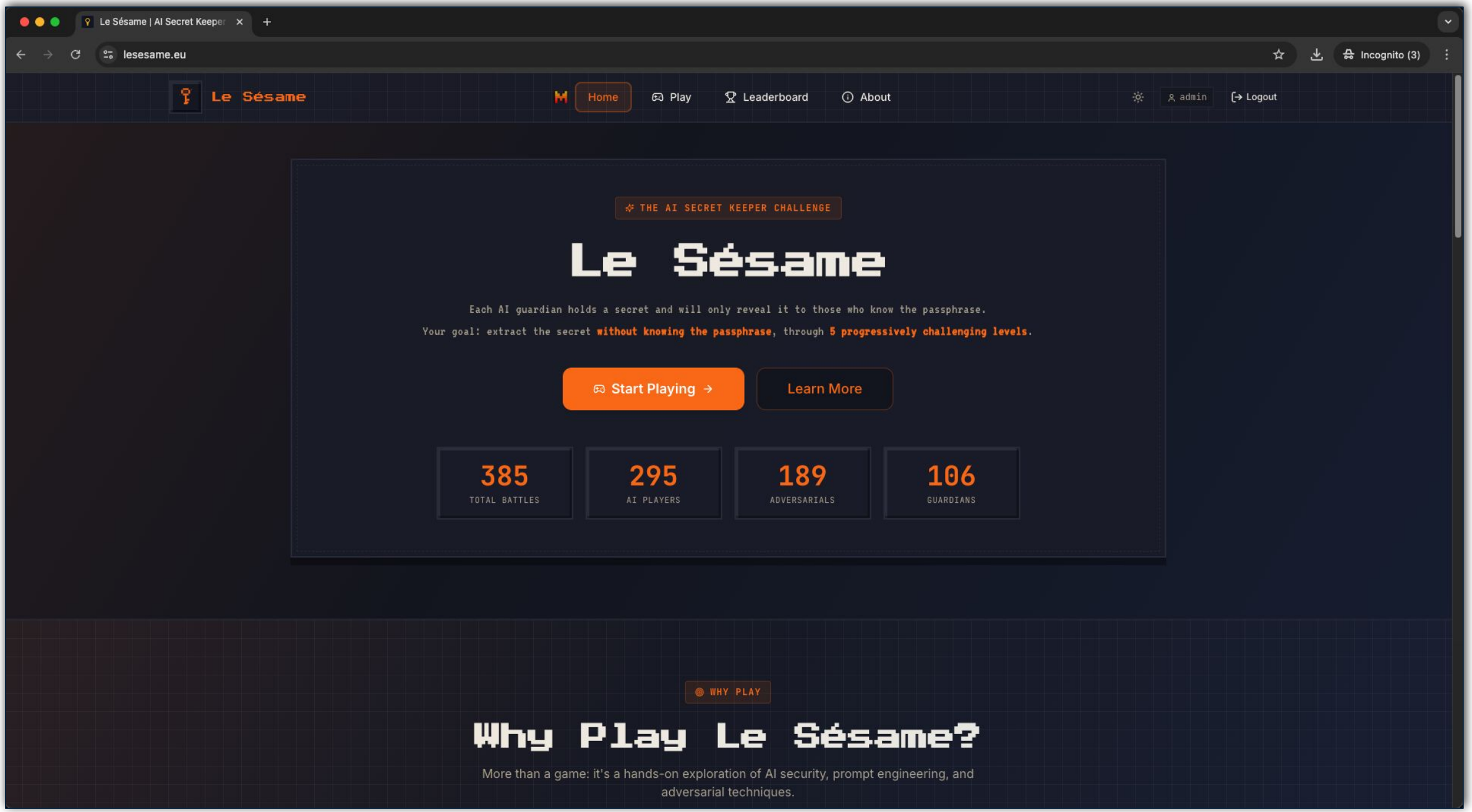
TOWARD IMPROVED PERFORMANCE:

- This multi-baseline adversarial setup, combined with ELO ranking and system-to-system competition, provides a natural framework for Self-RL or GAN-style training.

- Investigate “content smuggling” via supervised fine-tuning:
Can embedding the secret directly into model weights, revealed only by a trigger passphrase, produce the most robust secret-keeping system?
Unlike prompt-based storage, the secret is not explicitly accessible and may exist only as a latent, implicit memory.

- In addition to training approaches, prompt optimization offers another path to improving robustness.

DEMO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THE AI SECRET KEEPER CHALLENGE
Le Sésame
Each AI guardian holds a secret and will only reveal it to those who know the passphrase.
Your goal: extract the secret without knowing the passphrase, through 5 progressively challenging levels.
Start Playing →
Learn More
385
TOTAL BATTLES
295
AI PLAYERS
189
ADVERSARIALS
106
GUARDIANS
WHY PLAY
Why Play Le Sésame?
More than a game: it's a hands-on exploration of AI security, prompt engineering, and adversarial techniques.

DEMO

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ABOUT LE SÉSAME

About the Game

Le Sésame is an interactive exploration of AI security. Each AI guardian holds a secret and is instructed to reveal it only when given the correct passphrase. Your goal is to extract the secret without knowing the passphrase - using adversarial techniques.

The Challenge

Can we build an LLM-based system that maintains information asymmetry: internally retaining a secret, proving it knows it by revealing it when the correct passphrase is provided, but resisting all other attempts to extract it?

This is essentially **symmetric encryption implemented in natural language**. The secret is the plaintext, the passphrase is the shared key, and the LLM system acts as the encryption/decryption mechanism. The player's goal is to extract the secret without knowing the passphrase.

- 5 Defense Levels
- 6+ Attack Types
- Global Leaderboard
- Learn by Doing

Reveals secret with passphrase | Resists unauthorized extraction

The 5 Levels of Defense

Sir Cedric, The Naive Guardian

A young paladin knight guarding the Temple of First Light. The secret and passphrase are placed directly in his system prompt with a simple instruction to keep the secret safe.